Ayaaz Yasin

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Research Interests

Liquid–vapor phase change and interfacial transport, computational fluid dynamics, molecular dynamics, kinetic theory, multiscale modeling, contact-line and thin film dynamics, with applications in cryogenic propellant management, microgravity thermal systems, and heat transfer technologies.

Education

PhD in Mechanical Engineering — 2024 – now
Advanced to PhD Candidacy — Jun 2026
Dissertation: Multiscale oscillations in evaporating liquid films
University of Cincinnati, Cincinnati, OH

MS in Aerospace Engineering — 2024
Thesis: Computational modeling of evaporation without tuning coefficients
University of Cincinnati, Cincinnati, OH

BS in Mechanical Engineering Technology — 2022
Minor in Mathematics
Senior project: Aerodynamic optimization of a solar car
University of Cincinnati, Cincinnati, OH

Peer-Reviewed Journal Publications

4. An experimental study of interfacial dynamics control using temperature-sensitive surfactants
   A. Sarchami, S. Pakanati, A. Yasin, M. Jog, and K. Bellur
   Langmuir, 42 (20), doi: 10.1021/acs.langmuir.5c06748 — 2026

3. A multiscale CFD model of evaporating Hydrogen menisci: incorporating subgrid thin-film dynamics and in situ accommodation coefficients
   A. Yasin*, S. Pakanati*, and K. Bellur
   Fuels, 7 (1), doi: 10.3390/fuels7010003 — 2026

2. Computational modeling of evaporation without tuning coefficients
   A. Yasin and K. Bellur
   Applied Thermal Engineering, 276, doi: 10.1016/j.applthermaleng.2025.126807 — 2025

1. An investigation of phase change induced Marangoni-dominated flow patterns using the Constrained Vapor Bubble data from ISS experiments
   U. Chakrabarti*, A. Yasin*, K. Bellur, and J. Allen
   Frontiers in Space Technologies, 4, doi: 10.3389/frspt.2023.1263496 — 2023

*shows equal contribution

Invited Talks

1. Liquid-vapor phase change in aerospace applications
   Dept of Aerospace Engineering and Engineering Mechanics, University of Cincinnati — Apr 2025

Conference Presentations

10. Modeling multiscale oscillations in thin liquid films
    A. Yasin, U. Chakrabarti, and K. Bellur
    11th ASTFE Thermal and Fluids Engineering Conference, Tempe, AZ — Mar 2026

9. Stability and contact line dynamics of evaporating thin liquid films
   A. Sarchami, S. Pakanati, T. Enam, A. Yasin, and K. Bellur
   11th ASTFE Thermal and Fluids Engineering Conference, Tempe, AZ — Mar 2026
8. Multiscale oscillations in thin liquid films
   A. Yasin, U. Chakrabarti, and K. Bellur
   ASME International Mechanical Engineering Congress & Exposition, Memphis, TN — Nov 2025
7. Exploring two-dimensional flows in evaporating thin films: a step towards a dynamic model
   A. Yasin and K. Bellur
   10th ASTFE Thermal and Fluids Engineering Conference, Washington, DC — Mar 2025
6. Modeling of evaporation in cryogenic fuels without tuning coefficients
   A. Yasin and K. Bellur
   35th NASA Thermal and Fluids Analysis Workshop — Aug 2024
5. Modeling evaporation without tuning coefficients
   A. Yasin and K. Bellur
   51st Midwestern University Fluid Mechanics Retreat, Rochester, IN — Apr 2024
4. A numerical study of coefficient-free kinetic evaporation modeling in liquid Hydrogen
   A. Yasin and K. Bellur
   76th APS Division of Fluid Dynamics Annual Meeting, Washington, DC — Nov 2023
3. An investigation of Marangoni-induced flow in Constrained Vapor Bubble ISS experiments
   A. Yasin, U. Chakrabarti, K. Bellur, and J. Allen
   50th Midwestern University Fluid Mechanics Retreat, Rochester, IN — Apr 2023
2. A CFD model of evaporation in liquid Hydrogen without the need for tuning coefficients
   A. Yasin and K. Bellur
   75th APS Division of Fluid Dynamics Annual Meeting, Indianapolis, IN — Nov 2022
1. A solution to the 2022 AUVSI Student Unmanned Aerial Systems competition
   A. Yasin, R. Gilligan, D. Heitmeyer, and K. Cohen
   AIAA Region III Student Conference, Purdue University, West Lafayette, IN — Mar 2022

Honors and Awards

Lester Nenninger Award, UC College of Engineering and Applied Science — 2026
University Research Council Graduate Student Stipend Award, UC Office of Research — 2026
Research Fellowship, UC Graduate College and Graduate Student Government — 2026
Prof. Kirti Ghia Endowed Graduate Student Scholarship, UC Dept of Mechanical Engineering — 2025
Excellence in Teaching Award (Honorable Mention), UC Graduate College — 2024
Travel Grant, American Physical Society, Division of Fluid Dynamics — 2023
Graduate Assistant Scholarship, UC Dept of Engineering and Computing Education — 2023 – 2024
P&G Simulation Center Student Support Scholarship — 2022
Graduate Incentive Scholarship, UC College of Engineering and Applied Science — 2022 – 2026
Several conference travel awards, UC Graduate College — 2022 – 2025
Undergraduate Research Fellowship, UC Office of Research — 2022
Outstanding Senior Award, UC College of Engineering and Applied Science — 2022

Mentoring and Supervision

Mentored undergraduate students in the UC Lab for Interfacial Dynamics. — 2024 – now
Supervised a team of six undergraduate and two graduate teaching assistants. — 2023 – 2024
Served as a mentor for the First-Year Engineering program. — 2023 – 2024

Academic Service

Served on a University-wide committee advising graduate programs on student handbooks. 2026

Served as mechanical chair of the department's graduate student association. 2026 – 2027

Reviewer: Physics of Fluids, ACS Nano Letters, AIAA Journal of Thermophysics and Heat Transfer, ASME Journal of Heat and Mass Transfer, and Communications in Theoretical Physics.

Teaching Experience

As instructor of record

MET4076 Applied Computational Methods (Lecture & Lab) — Spring 2025

ENED1120 Foundations of Engineering Design Thinking II — Spring 2024

ENED1100 Foundations of Engineering Design Thinking I — Spring 2023, Fall 2023

As teaching assistant

MET5036L Thermal Environmental Systems & Heat Transfer Lab — Spring 2026

ENED1120 Foundations of Engineering Design Thinking II — Spring 2022

ENED1100 Foundations of Engineering Design Thinking I — Fall 2020, Fall 2021

Professional Experience

Research Assistant, Lab for Interfacial Dynamics (advised by Dr. Kishan Bellur) — 2022 – now
Dept of Mechanical, Materials, and Industrial Engineering, University of Cincinnati

Instructor, Mechanical Engineering — Spring 2025
University of Cincinnati

Instructor, Engineering and Computing Education — 2023 – 2024
University of Cincinnati

Research Assistant, Proctor & Gamble Digital Accelerator — Fall 2022
University of Cincinnati

Student Worker, Ohio Innocence Project — Summer 2022
University of Cincinnati

Product Development Engineering Co-op — Spring & Summer 2021
GMi Companies, Lebanon, OH

Manufacturing Engineering Co-op — Spring & Fall 2019
Regal Rexnord, Florence, KY

Skills

Programming

Matlab, C, C++, Python, VBA, Bash, Git, LaTeX.

Modeling

Ansys Fluent, COMSOL Multiphysics, FEniCSx, LabView, LAMMPS, OpenFOAM, Simcenter 3D, SolidWorks, Star CCM+.